Cemetery Management System

Proposal

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Background

Since South Africa was first inhabited, people have been burying their dead following traditional customs and practices. Over the past two centuries funerals and burial sites and their management have evolved. As city centers emerged, management, maintenance, record-keeping and environmental issues related to cemetery management became key issues of concern. People are increasingly migrating into urban areas and this development puts strain on the management of limited municipal resources. Municipalities need to both emulate and create good practices in cemetery management.

Cemeteries in South African urban areas have historically been planned as spacious, landscaped gardens to house human remains. However, older cemeteries have run, or are running out of space for new burials, especially in major urban municipalities. This is as a result of rapid urbanization in towns and cities, which results in fierce competition for well-located land.

Municipalities are also faced with the challenge of cemeteries developing informally or on land geographically unsuitable for in-ground burial. Among most cultures in South Africa, the cemeteries are more than just a place of burial. They carry spiritual, religious and cultural significance.

The main challenge facing cemetery management today is striking an acceptable balance between spiritual, religious and cultural practices and the present and future availability of land for cemeteries. To address this challenge, some municipalities have implemented innovative practices which could be inculcated into the norms, standards, by-laws and electronic software systems of cemetery management. Record-keeping is a very important function in managing cemeteries.

Electronic record-keeping is ideal, yet not widely used outside the major municipalities. This is where Electronic Cemetery Management System would come in to alleviate some of the challenges experienced in the management of graveyards.

Introduction

Local Municipalities has identified a need to develop an Electronic Cemetery Management System which will assist the municipality to manage the graves and burial process effectively. The need for developing the Municipal Electronic Cemetery Management System is informed by losing previous grave and burial process information due to deteriorating paper records. The additional need is created by the reality that the current paper recording system does not assist the municipality to monitor remaining capacity and life span of a municipal cemeteries.

The Electronic Cemetery Management System gives municipalities and managers of cemeteries a simple, web-based system for access to and management of cemetery data and the registry of the deceased such as: -.

a) Record, store and manage burial, cremation and exhumation information
b) Electronic archiving of historical burial and grave records based on data gathered throughout the day-to-day operations of municipal cemeteries, as well as through the conversion based historical of paper- burial records into electronic format
c) The information recorded includes deceased details, grave details, GPS coordinates to the graveyard, next of kin details, payment details and undertaker details
d) Dashboard showing sales statistics, user statistics, graveyard statistics, racial statistics, undertaker statistics and visual map showing all graveyards, together with their statistics
e) Audit log of any action happening on the system
f) Report showing graveyards/crematoriums and number of interments recorded
g) Report showing users and number of interments that were recorded by them
h) Attaching supporting documents to the interment records like death certificate, deceased's identity document, medical report etc.
i) Facility for the public to see grave/cemetery related notes, download burial application forms and other cemetery related document such as By-Laws, Tariffs, Act, Policies and Norms and Standards
j) Allow public to search grave information using any Internet-connected device 365/24/7, from anywhere around the world
k) Send grave application approval and grave number to the applicant/next of kin via SMS and/or email. Mode of communication depends on the availability of contact information at the time of application
l) Process a predated raw data which was manually captured by the municipality
m) Automatically compile read-only electronic audit trail of all user activities; date, time & user stamp

Problem Statement

- Municipality is at risk of **non-adherence to legislation**, especially in terms of records keeping and Funeral, Burial and Cremation Services Act 2002
- The **intellectual property** of the municipality is not under the control of the organization
- **Paper-based documents** pose security risks
- Ever **increasing physical storage space** is costly and unsustainable.
- **Providing information** is slow and untrustworthy
- **Documents are not shared** internally as well as intra-local government and **lots of duplication** occurs.

Solution to the Problem

Record Capturing

Interment or burial is a method of final disposition wherein a deceased person or animal is placed into the ground, sometimes with objects. This is usually accomplished by excavating a pit or trench, placing the deceased and objects in it, and covering it over. A funeral is a ceremony that accompanies the final disposition. Humans have been burying their dead since shortly after the origin of the species. Burial is often seen as indicating respect for the dead. It has been used to prevent the odor of decay, to give family members closure and prevent them from witnessing the decomposition of their loved ones, and in many cultures it has been seen as a necessary step for the deceased to enter the afterlife or to give back to the cycle of life.

In this module the information recorded includes deceased details, grave details, GPS coordinates to the graveyard and actual location of the grace, next of kin details, payment details and undertaker details.

Add New

Show 10 entries Search:

Data Capturer	Deceased	Date of Birth	Date of Death	Date of Burial	Grave Number	Status	
Anna Masithulela	Jonker Coetzee	12 Sep 1938	03 Apr 2008	10 Apr 2008	319	Finalised	View
Anna Masithulela	Uhone Makhadzi Maitakhole	18 Nov 1994	09 Dec 2015	15 Dec 2015	214	Finalised	View
Anna Masithulela	Rullet Helms	22 May 1968	15 Jul 2017	22 Jul 2017	358	Finalised	View
Anna Masithulela	Violet Legae	13 Nov 1954	20 Jan 2019	26 Jan 2019	309	Finalised	View
Anna Masithulela	Rulani Arinao	21 May 1985	04 Aug 2018	11 Aug 2018	242	Finalised	View

Figure 1: Capturing Interment information

The above screen shows the names of the Data Capture and specific record they captured of the deceased, their date of birth, the date they departed and the date when they are buried, the grave number and finally the application status weather it was Finalized or Captured.

Cemetery staff in the municipality are committed to providing excellent burial services and maintenance, and to providing sensitive customer service to families who are going through the process of planning a funeral

Interments Report

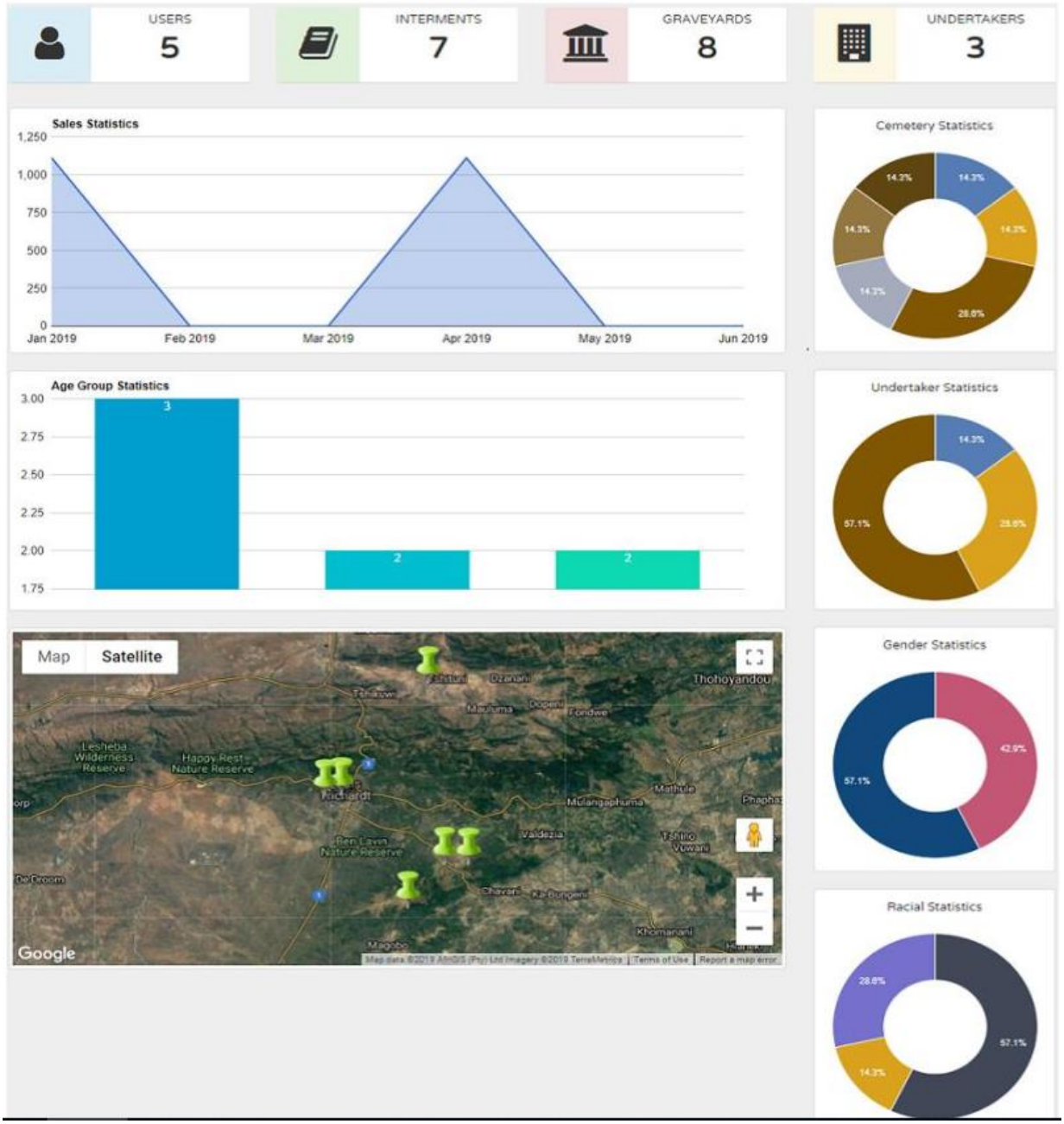
Start Date 2019/06/06 End Date 2019/06/11 Search Query Enter name, surname or ID number Search

Grave site	Deceased	Date of Birth	Date of Death	Date of Burial	Grave Number	Status
Waterval New Cemetery	Jonker Coetzee	12 Sep 1938	03 Apr 2008	10 Apr 2008	319	Finalised
Vleifontein Cemetery	Uhone Makhadzi Maitakhole	18 Nov 1994	09 Dec 2015	15 Dec 2015	214	Finalised
Louis Trichardt Cemetery Site	Rullet Helms	22 May 1968	15 Jul 2017	22 Jul 2017	358	Finalised
Louis Trichardt Cemetery Site	Violet Legae	13 Nov 1954	20 Jan 2019	26 Jan 2019	309	Finalised
Tshikota Cemetery Site	Rulani Arinao	21 May 1985	04 Aug 2018	11 Aug 2018	242	Finalised
Hindu Crematorium Site	Ian Naidoo	11 Jun 1988	11 Jun 2018	18 Jun 2018	365	Finalised
Dzanani Cemetery	Setshege Makofane	23 Jun 1980	09 Apr 2019	15 Apr 2019	236	Finalised

Figure 2: Interment information Report

Administration Page

Admin page is where you manage your cemetery details, users, interments, undertaker's etc. As an administrator/user, when you login, you will be presented with a screen shown below.



Graveyards Management

Electronic Cemetery management includes many skill sets. These range from the most hands-on tasks on the ground, from grave-digging to landscaping and horticulture, to the more high-level tasks undertaken by managers, including the scheduling and logistics associated with running a cemetery. On a day-to-day and weekly basis, the cemetery officer and horticulturalist have to plan for the funerals that have been booked in their cemetery, taking care of everything from the booking of the grave and burial method, to the digging of the grave and making sure each burial goes smoothly. They also have to manage the flow of visitors to the cemetery paying their respects at the graves of loved ones, and perhaps unveiling tombstones, and ensure their security and the maintenance and upkeep of the cemetery.

The cemetery officer informs the cemetery manager for the municipality of the logistics at the grave site itself; the number of funerals that have been booked through undertakers, and whether they are for first burials, or reburials and other categories. Based on this information the head gardener decides how many graves should be dug and confirms this figure in consultation with the cemetery manager. Gardeners are responsible for maintenance and digging the holes for graves. In the past this used be done by hand, but it is now done using a tractor with a backhoe.

Cemetery maintenance covers a lot of different areas. The access road, internal roads and internal walkways either need to be tarred or, if dirt, then graded and kept smooth and passable in all weathers. Fences and gates, and building structures must be maintained, as well as signage and grave numbers. Horticultural issues include mowing of the lawns, weed control, watering, and pruning trees, shrubs and flowers. Finally, water, sewer and storm-water drains must be kept in good working order. Cemetery management is a profession that is not often spoken of, but it is so essential in one of the most important rites of life, the final goodbye.

In this module the system will be used to add new grave details, view, edit, and delete existing graveyard details.

Add New

Graveyards

Show 10 entries

Search:

Region	Caretaker	Name	Latitude	Longitude	
Dzanani Region		Dzanani Cemetery	-22.889513	30.029809	View Edit
Makhado Region		Hindu Crematorium Site	-23.05897	29.900911	View Edit
Makhado Region		Louis Trichardt Cemetery Site	-23.058097	29.899456	View Edit
Makhado Region		Muslim Crematorium Site	-23.058715	29.900906	View Edit
Makhado Region		Tshikota Cemetery Site	-23.060353	29.876122	View Edit
Waterval Region		Vleifontein Cemetery	-23.231771	29.999214	View Edit
Waterval Region		Waterval New Cemetery	-23.165487	30.090808	View Edit
Waterval Region		Waterval Old Cemetery	-23.164667	30.054155	View Edit

Example – Add New Graveyard details

Undertakers Management

A funeral director, also known as an undertaker (British English) or mortician (American English), is a professional involved in the business of funeral rites. These tasks often entail the embalming and burial or cremation of the dead, as well as the arrangements for the funeral ceremony (although not the directing and conducting of the funeral itself unless clergy are not present). Funeral directors may at times be asked to perform tasks such as dressing (in garments usually suitable for daily wear), casketing (placing the human body in the coffin), and cossetting (applying any sort of cosmetic or substance to the best viewable areas of the corpse for the purpose of enhancing its appearance). A funeral director may work at a funeral home or be an independent employee.

In this sub-module, the details of an undertaker are captured, edited, deleted and managed according to the desire of the municipality.

AVBOB Funeral Parlour

Name	AVBOB Funeral Parlour
Contact Number	015 516 0801
Email Address	

Edit Delete

Example – Edit Undertaker details

Tariffs Management

Budgets of municipalities in South Africa are financed through own revenue sources and fiscal transfers. Own revenue includes a variety of sources such as property tax, service charges and interests earned on investments. However, other sources like allocation and management of gravesites should considered and managed properly, currently municipalities are using manual processes which are tedious and error prone. This sub-module assists the municipality in management of tariffs for cemetery services rendered by the municipality.

Add New

Tariffs

Show 10 entries Search:

Title	Resident		Non-Resident		
	Adult	Minor	Adult	Minor	
Reservation of grave	R526,80	R332,40	R1 047,70	R664,80	View Edit
Purchase of grave	R526,80	R332,40	R1 047,70	R664,80	View Edit
Opening and closing of grave	R444,90	R221,30	R888,50	R468,30	View Edit
Widening and deepening of grave	R138,40	R138,40	R278,10	R278,10	View Edit
Use of niche	R528,30	R528,30	R1 053,50	R1 053,50	View Edit
Burial of pauper	R0,00	R0,00	R0,00	R0,00	View Edit

Example – Tariff Management Screen